ENGR 103: Engineering Computation and Algorithmic Thinking

Homework Assignment #4 – Learning Curves

Part 1 – Design

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Step 1 – Understanding the Problem (7.5 pts)
Use the spaces provided below to answer the following questions:

- (1.5 pts) Do you understand everything in the problem? List anything you do not fully understand.

Yes, I understand everything.

- (2 pts) What are the functional requirements of the program, i.e., what does it need to do?

The program needs to take input values for the first cycle time, the desired cycle time, and the slope, while checking for errors in the input, and calculate the cycle time for each cycle and detect when the cycle time first falls to or below the goal time, and report the learning percent when the goal is achieved.

- (2 pts) What assumptions are you making?

The decrease in time to perform a task.

- (2 pts) What are the inputs, outputs, etc.?

Inputs:

1. First cycle time
2. Desired cycle time
3. Slope of decrease (n)

Outputs:

1. Cycle number (each iteration)
2. Cycle time (each iteration)
3. When the cycle time is less than or equal to the desired, and the learning percent
    1.

Step 2 – Devise a Plan (10 pts)

Use the space provided below to answer the following questions:

- (1.5 pts) What are the decisions that need to be made in this program?

How to gather input

How to ensure only numbers in the correct range are input

How to calculate actual values

How to check when to end the loop

- (2 pts) What is the sequence of steps you need to complete?

Get input values

Calculate slope

Loop until time is at or below the desired value

Print results

- (1.5 pts) How are you going to do the required calculations?

Gather inputs, and put them into the $n = \Delta y / \Delta x$, then use that to loop through the rest in the

$Y = kx^n$ equation.

(5 pts) Based on your answers to the questions in Step 2, provide an algorithm as **pseudocode** or as a **flowchart** showing the specific steps that are needed to create the required computer program. **Be very explicit!!!**

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Input first cycle time

Input desired cycle time

Input slope of decrease (n)

Make sure each input is correct when it is entered


While loop

    For 100

        Print current time

        If goal time is reached print that and kick out of both loops

    If time not reached check if the user wants to continue
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<u>Step 4a – Looking Back (7.5 pts)</u>

Create a test plan (on paper) with test cases (i.e., good, bad, and edge cases).

- What are the good, bad, and edge test cases for ALL input in the program?
- What do you expect the results to be in each test case?

Use the table shown below to organize your proposed test cases. Test Case #1 is provided only as an example. You need to provide <u>at least five more test cases</u>. Note: Your test cases must incorporate the values of numerical inputs and numerical results your program will request and/or produce.

If you want to see more test case examples, consult the documents included in the module titled "POLYA 4SDP AND PYTHON STYLE GUIDELINES" in the course's Canvas website.

Test Case #	Value(s)	Expected	Good or Bad?
1	*n = 1,* *test score = 100*	*The average test score should be 100.*	*Good*
2			
3			
4			
5			
6			
7			
8			
9			